Inference with weakly dependent data II

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Plan for this slide deck

1. New Keynesian Phillips curve
2. IV regression with HAC standard errors
3. Weak instruments
4. Consumption Euler equation

Phillips curve: prediction vs. causality

- The Phillips curve we estimated in the previous lecture was a predictive relationship.
- There is no reason to believe any correlation (or lack thereof) would be causal.
- Inflation and unemployment could be correlated simply because they are both caused by the same underlying economic shocks (e.g., monetary policy). When unemployment rises/falls, we're not holding everything else constant.
- In this lecture, we will discuss how to estimate a more modern, causal version of the Phillips curve, guided by economic theory.

Nominal rigidities

- Many modern theoretical macro models assume that firms face **nominal rigidities** when setting their prices.
- This means that they do not optimally change their prices every period, even when business conditions change.
  - Could reflect inattention, physical costs of changing prices ("menu costs"), or fear of upsetting customers.
- In periods when firms set their price optimally, they do so based on expectations of business conditions over the next several periods (since they may not get to re-optimize prices again for a while).
- In periods where firms do not set an optimal price, they might passively pro-rate their price using past inflation.

New Keynesian Phillips curve

- Due to the nominal rigidities, these models often imply a structural/causal equation of the following form, called the **new Keynesian Phillips curve** (NKPC):

$$\pi_t = c + \gamma_f E_t[\pi_{t+1}] + \gamma_b \pi_{t-1} + \lambda \chi_t + u_t.$$

  - π_t: inflation
  - $E_t[\pi_{t+1}]$: expected next-period inflation
  - χ_t: measure of economic slack (say, the unemployment rate)
  - u_t: cost-push shock (e.g., oil price shock)

- γ_f and γ_b measure the extent of forward-looking or backward-looking price setting behavior.

- λ (the slope of the NKPC) measures the feedback from the real economy to inflation.

NKPC: estimation

$$\pi_t = c + \gamma_f E_t[\pi_{t+1}] + \gamma_b \pi_{t-1} + \lambda \chi_t + u_t$$

- We would like to estimate the parameters of this equation, as they are important inputs into economic policy.
- But we can't just run an OLS regression of inflation on the right-hand-side variables, for two reasons:
  1. We do not directly observe the expectation $E_t[\pi_{t+1}]$. (Firms may make different forecasts from households or professional forecasters.)
  2. The cost-push shock u_t may be correlated with $E_t[\pi_{t+1}]$ and χ_t, so these covariates are endogenous.
- We'll have to use a different estimation approach.

NKPC: estimation (cont.)

$$\pi_t = c + \gamma_f E_t[\pi_{t+1}] + \gamma_b \pi_{t-1} + \lambda \chi_t + u_t$$

- Define the one-step-ahead forecast error $\eta_{t+1} \equiv \pi_{t+1} - E_t[\pi_{t+1}]$.
- Substitute $E_t[\pi_{t+1}] = \pi_{t+1} - \eta_{t+1}$ in the NKPC:

$$\pi_t = c + \gamma_f \pi_{t+1} + \gamma_b \pi_{t-1} + \lambda \chi_t + v_t, \quad \text{where} \quad v_t \equiv u_t - \gamma_f \eta_{t+1}.$$

- We make two key assumptions:
  1. The cost-push shock u_t is unpredictable by past data: $E[u_t z_{t-\ell}] = 0$ for $\ell \geq 1$, where $\{z_t\}$ is any macro time series.
  2. Firms' expectations are **rational**, so the forecast errors are unpredictable by past data: $E[\eta_{t+1} z_{t-\ell}] = 0$ for $\ell \geq 0$.
- Exercise: show these assumptions imply $E[v_t z_{t-\ell}] = 0$ for all $\ell \geq 1$.

NKPC: estimation (cont.)

$$\pi_t = c + \gamma_f \pi_{t+1} + \gamma_b \pi_{t-1} + \lambda \chi_t + v_t,$$

$$E[v_t z_{t-\ell}] = 0 \quad \text{for all} \quad \ell \geq 1$$

- In summary: though the covariates π_{t+1} and χ_t may not be uncorrelated with v_t, any lagged variable $z_{t-\ell}$ is.
- This suggests using lagged variables as **instrumental variables** (IVs).
- Let's review IV regression and think about how it can be adapted to time series data...

IV regression

- Consider a structural outcome equation

$$y_t = \beta_0 + \beta_1 x_{1t} + \cdots + \beta_k x_{kt} + v_t.$$

- We observe r instruments $z_{1t}, \ldots, z_{rt}$ and assume the **exogeneity conditions**

$$E[v_t] = 0 \quad \text{and} \quad E[z_{jt} v_t] = 0, \text{ for } j = 1, \ldots, r.$$

- In vector notation:

$$y_t = \boldsymbol{x}_t' \boldsymbol{\beta} + v_t, \quad E[\boldsymbol{z}_t v_t] = \mathbf{0}, \quad \text{where}$$

$$\underset{(k+1)\times 1}{\boldsymbol{\beta}} \equiv \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \underset{(k+1)\times 1}{\boldsymbol{x}_t} \equiv \begin{pmatrix} 1 \\ x_{1t} \\ \vdots \\ x_{kt} \end{pmatrix}, \quad \underset{(r+1)\times 1}{\boldsymbol{z}_t} \equiv \begin{pmatrix} 1 \\ z_{1t} \\ \vdots \\ z_{rt} \end{pmatrix}.$$

Example: NKPC

$$y_t = \boldsymbol{x}_t'\beta + v_t, \quad E[\boldsymbol{z}_t v_t] = \mathbf{0}$$

$$y_t = \pi_t, \quad \boldsymbol{x}_t = \begin{pmatrix} 1 \\ \pi_{t+1} \\ \chi_t \\ \pi_{t-1} \end{pmatrix}, \quad \boldsymbol{z}_t = \begin{pmatrix} 1 \\ \pi_{t-1} \\ \chi_{t-1} \\ \pi_{t-2} \\ \chi_{t-2} \\ \pi_{t-3} \\ \chi_{t-3} \\ \vdots \end{pmatrix}$$

($\boldsymbol{z}_t$ could also include lags of variables other than π_t and χ_t.)

IV regression: identification

$$y_t = \boldsymbol{x}_t'\beta + v_t, \quad E[\boldsymbol{z}_t v_t] = \mathbf{0}$$

- Notice that

$$E[\boldsymbol{z}_t y_t] = E[\boldsymbol{z}_t(\boldsymbol{x}_t'\beta + v_t)] = E[\boldsymbol{z}_t \boldsymbol{x}_t']\beta + E[\boldsymbol{z}_t v_t] = E[\boldsymbol{z}_t \boldsymbol{x}_t']\beta.$$

- So if $\mathbf{\Pi}$ is a $(k+1) \times (r+1)$ matrix,

$$\mathbf{\Pi} E[\boldsymbol{z}_t y_t] = \mathbf{\Pi} E[\boldsymbol{z}_t \boldsymbol{x}_t']\beta \quad \Longrightarrow \quad \beta = \left(\mathbf{\Pi} E[\boldsymbol{z}_t \boldsymbol{x}_t']\right)^{-1} \mathbf{\Pi} E[\boldsymbol{z}_t y_t],$$

  provided that the $(k+1) \times (k+1)$ matrix $\mathbf{\Pi} E[\boldsymbol{z}_t \boldsymbol{x}_t']$ is non-singular.

2SLS

$$\beta = \left(\mathbf{\Pi} E[\boldsymbol{z}_t \boldsymbol{x}_t']\right)^{-1} \mathbf{\Pi} E[\boldsymbol{z}_t y_t]$$

- Conventional choice:

$$\mathbf{\Pi} \equiv E[\boldsymbol{x}_t \boldsymbol{z}_t'] \left(E[\boldsymbol{z}_t \boldsymbol{z}_t']\right)^{-1}.$$

- Then $\mathbf{\Pi} E[\boldsymbol{z}_t \boldsymbol{x}_t']$ is non-singular if and only if the $(r+1) \times (k+1)$ matrix $E[\boldsymbol{z}_t \boldsymbol{x}_t']$ has full column rank (the **relevance condition**).
  - Necessary condition 1: $r \geq k$.
  - Necessary condition 2: each of the covariates $x_{1t}, \ldots, x_{kt}$ must be at least somewhat predictable by the IVs $z_{1t}, \ldots, z_{rt}$.
- Suggests the **two-stage least squares** (2SLS) estimator

$$\hat{\beta}_{\text{2SLS}} \equiv \left(\hat{\mathbf{\Pi}} \sum_{t=1}^{T} \boldsymbol{z}_t \boldsymbol{x}_t'\right)^{-1} \hat{\mathbf{\Pi}} \sum_{t=1}^{T} \boldsymbol{z}_t y_t, \text{ where } \hat{\mathbf{\Pi}} \equiv \sum_{t=1}^{T} \boldsymbol{x}_t \boldsymbol{z}_t' \left(\sum_{t=1}^{T} \boldsymbol{z}_t \boldsymbol{z}_t'\right)^{-1}.$$

2SLS (cont.)

$$\hat{\boldsymbol{\beta}}_{\text{2SLS}} \equiv \left(\hat{\boldsymbol{\Pi}} \sum_{t=1}^{T} \boldsymbol{z}_t \boldsymbol{x}_t' \right)^{-1} \hat{\boldsymbol{\Pi}} \sum_{t=1}^{T} \boldsymbol{z}_t y_t, \quad \hat{\boldsymbol{\Pi}} \equiv \sum_{t=1}^{T} \boldsymbol{x}_t \boldsymbol{z}_t' \left(\sum_{t=1}^{T} \boldsymbol{z}_t \boldsymbol{z}_t' \right)^{-1}$$

- Recall from a previous econometrics course that the estimator can be computed in two stages (hence the name):
  1. For each covariate $j = 1, \ldots, k$: run the **first-stage** OLS regression of x_{jt} on $z_{1t}, \ldots, z_{rt}$; store the fitted values $\{\hat{x}_{jt}\}_{t=1}^{T}$.
     - The coefficients in the j-th first-stage regression are the elements in the j-th row of $\hat{\boldsymbol{\Pi}}$.
  2. Run the **second-stage** OLS regression of y_t on $\hat{x}_{1t}, \ldots, \hat{x}_{kt}$; the coefficients are the elements of $\hat{\boldsymbol{\beta}}_{\text{2SLS}}$.
- In practice, we make software do all this for us.

2SLS: asymptotic distribution

- Consider the case $k = r = 1$ with no intercept:

$$y_t = \beta x_t + v_t, \quad E[z_t v_t] = 0.$$

- Then

$$\hat{\beta}_{2\text{SLS}} = \frac{\sum_{t=1}^{T} y_t z_t}{\sum_{t=1}^{T} x_t z_t} = \beta + \frac{\sum_{t=1}^{T} v_t z_t}{\sum_{t=1}^{T} x_t z_t}$$

$$\Longrightarrow \quad \sqrt{T}(\hat{\beta}_{2\text{SLS}} - \beta) = \frac{\frac{1}{\sqrt{T}} \sum_{t=1}^{T} v_t z_t}{\frac{1}{T} \sum_{t=1}^{T} x_t z_t}.$$

- Assuming weakly dependent data, the numerator satisfies a CLT and the denominator a LLN.

2SLS: asymptotic distribution (cont.)

- So the 2SLS estimator is approximately normal in large samples:

$$\sqrt{T}(\hat{\beta}_{2SLS} - \beta) = \frac{\frac{1}{\sqrt{T}}\sum_{t=1}^{T} v_t z_t}{\frac{1}{T}\sum_{t=1}^{T} x_t z_t} \xrightarrow{d} N\left(0, \frac{\text{LRV}(v_t z_t)}{(E[x_t z_t])^2}\right).$$

- Standard errors:

$$SE(\hat{\beta}_{2SLS}) = \sqrt{\frac{1}{T} \times \frac{\widehat{LRV}}{\left(\frac{1}{T}\sum_{t=1}^{T} x_t z_t\right)^2}},$$

  where $\widehat{LRV}$ is a HAC estimator applied to $\{\hat{v}_t z_t\}$, with $\hat{v}_t \equiv y_t - \hat{\beta}_{2SLS} x_t$.

- These arguments can be extended to the case with general $k \leq r$. In practice, statistical software packages have built-in 2SLS routines.

Let's estimate the NKPC!

```
from linearmodels.iv import IV2SLS

# Load data
df_cpi = pd.read_csv('data/CPIAUCSL.csv', index_col='observation_date',
    parse_dates=True)
df_unemp = pd.read_csv('data/UNRATE.csv', index_col='observation_date',
    parse_dates=True)

# Merge and interpolate missing values in Oct 2025
df = df_cpi.join(df_unemp, how='inner').interpolate()

# Compute annualized month-over-month inflation
df['infl'] = 1200 * np.log(df['CPIAUCSL']).diff()
df = df.dropna()
```

```
# Prepare variables
nlag_iv = 3

df_iv = df.assign(
    infl_lead=df['infl'].shift(-1),
    **{f'infl_lag{l}':   df['infl'].shift(l)   for l in range(1, nlag_iv + 1)},
    **{f'UNRATE_lag{l}': df['UNRATE'].shift(l) for l in range(1, nlag_iv + 1)},
).dropna()
df_iv = sm.add_constant(df_iv)

instr_cols = [f'infl_lag{l}'   for l in range(2, nlag_iv + 1)] \
           + [f'UNRATE_lag{l}' for l in range(1, nlag_iv + 1)]

# IV regression
T = len(df_iv)
nw_bw = int(np.ceil(0.75 * T**(1/3)))

result = IV2SLS(
    dependent=df_iv['infl'],
    exog=df_iv[['const', 'infl_lag1']],
    endog=df_iv[['infl_lead', 'UNRATE']],
    instruments=df_iv[instr_cols],
).fit(cov_type='kernel', bandwidth=nw_bw - 1)
print(result)
```

IV-2SLS Estimation Summary

Dep. Variable:	infl	R-squared:	0.4385
Estimator:	IV-2SLS	Adj. R-squared:	0.4366
No. Observations:	933	F-statistic:	1041.4
Date:	Wed, Apr 15 2026	P-value (F-stat)	0.0000
Time:	23:12:23	Distribution:	chi2(3)
Cov. Estimator:	kernel		

Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
const	-0.0199	0.1733	-0.1148	0.9086	-0.3596	0.3198
infl_lag1	0.2772	0.0947	2.9264	0.0034	0.0915	0.4628
infl_lead	0.7316	0.1554	4.7078	0.0000	0.4270	1.0363
UNRATE	-0.0016	0.0279	-0.0572	0.9544	-0.0563	0.0531

Endogenous: infl_lead, UNRATE
Instruments: infl_lag2, infl_lag3, UNRATE_lag1, UNRATE_lag2, UNRATE_lag3
Kernel Covariance (HAC)
Debiased: False
Kernel: bartlett
Automatic Bandwidth: False
Bandwidth: 7

Syntax

- Notice the terminology in the `IV2SLS` function:
  - `dependent`: y_t.
  - `exog`: variables in $\boldsymbol{x}_t$ that are also in $\boldsymbol{z}_t$ (includes at least the intercept).
  - `endog`: variables in $\boldsymbol{x}_t$ that are not in $\boldsymbol{z}_t$.
  - `instruments`: variables in $\boldsymbol{z}_t$ that are not in $\boldsymbol{x}_t$.
- Other software (e.g., Stata) uses similar terminology.

```
# Test the restriction gamma_f = gamma_b
print(result.wald_test(formula='infl_lead = infl_lag1'))
```

Result:

```
Linear Equality Hypothesis Test
H0: Linear equality constraint is valid
Statistic: 3.3525
P-value: 0.0671
Distributed: chi2(1)
```

Weak instruments

- When the relevance condition is close to failing, the 2SLS estimator may be very non-normally distributed in finite samples.
- Conventional SEs/CIs/tests can be highly inaccurate in this case.
- Intuitively, if the IVs collectively have little predictive power for some of the covariates, then there is no way to isolate sufficient exogenous variation in the latter to reliably identify causal effects.
- It's customary to diagnose weak IV problems with **first-stage F-statistics**:
  - In each first-stage regression for endogenous covariates, compute the F-stat for the hypothesis that all the IVs $\boldsymbol{z}_t$ (except those also included in $\boldsymbol{x}_t$) have coefficients $= 0$.
  - If F-stat < 10, the IVs are weak.
- Warning: this is far from foolproof. Just a "rule of thumb".

```
=== First stage: UNRATE ===
                            OLS Regression Results
==============================================================================
Dep. Variable:                 UNRATE   R-squared:                       0.942
...
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.1723      0.107      1.610      0.107      -0.037       0.382
infl_lag1     -0.0068      0.008     -0.822      0.411      -0.023       0.009
infl_lag2      0.0025      0.003      0.730      0.465      -0.004       0.009
infl_lag3      0.0043      0.005      0.915      0.360      -0.005       0.014
UNRATE_lag1    1.0141      0.063     16.145      0.000       0.891       1.137
UNRATE_lag2   -0.0691      0.076     -0.911      0.362      -0.218       0.079
UNRATE_lag3    0.0247      0.088      0.280      0.780      -0.148       0.198
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC)
using 7 lags and without small sample correction

F-stat on instruments (infl_lag2, infl_lag3, UNRATE_lag1, UNRATE_lag2, UNRATE_lag3):
<F test: F=1738.038020505514, p=0.0, df_denom=926, df_num=5>
```

```
=== First stage: UNRATE ===
                            OLS Regression Results
==============================================================================
Dep. Variable:                 UNRATE   R-squared:                       0.942
...
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.1723      0.107      1.610      0.107      -0.037       0.382
infl_lag1     -0.0068      0.008     -0.822      0.411      -0.023       0.009
infl_lag2      0.0025      0.003      0.730      0.465      -0.004       0.009
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<F test: F=1738.038020505514, p=0.0, df_denom=926, df_num=5>
```

Strong!

```
=== First stage: infl_lead ===
                            OLS Regression Results
==============================================================================
Dep. Variable:              infl_lead   R-squared:                       0.249
...
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.8169      0.429      1.903      0.057      -0.025       1.658
infl_lag1      0.3301      0.044      7.431      0.000       0.243       0.417
infl_lag2      0.1047      0.049      2.131      0.033       0.008       0.201
infl_lag3      0.1459      0.053      2.746      0.006       0.042       0.250
UNRATE_lag1    0.3956      0.254      1.559      0.119      -0.102       0.893
UNRATE_lag2   -0.2766      0.359     -0.772      0.440      -0.979       0.426
UNRATE_lag3   -0.0163      0.448     -0.036      0.971      -0.895       0.863
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity and autocorrelation robust (HAC)
using 7 lags and without small sample correction

F-stat on instruments (infl_lag2, infl_lag3, UNRATE_lag1, UNRATE_lag2, UNRATE_lag3):
<F test: F=4.960382068782446, p=0.00017281232129073444, df_denom=926, df_num=5>
```

```
=== First stage: infl_lead ===
                            OLS Regression Results
==============================================================================
Dep. Variable:              infl_lead   R-squared:                       0.249
...
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.8169      0.429      1.903      0.057      -0.025       1.658
infl_lag1      0.3301      0.044      7.431      0.000       0.243       0.417
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```

Weak!

Weak instruments: solutions

- One could look for different instrument sets with higher first-stage F-stats. But the fundamental problem for NKPC estimation is that we don't know many variables that Granger-cause inflation.
- Instead, there exist procedures for constructing confidence sets that are robust to the strength of the IVs.
- These **weak-IV-robust** procedures often lead to large confidence sets, but they could still be informative in some settings.
- Details on the reading list: Mavroeidis, Plagborg-Moller & Stock (2014); Andrews, Stock & Sun (2019).

Consumption Euler equation

- Many structural macro models also imply an optimality condition for aggregate consumption called the **consumption Euler equation**:

$$y_t = c + \delta_f E_t[y_{t+1}] + \delta_b y_{t-1} - \tau(i_t - E_t[\pi_{t+1}]) + u_t.$$

  - y_t: consumption (sometimes substituted with GDP)
  - i_t: nominal interest rate
  - π_t: inflation
  - u_t: demand shock

- $\delta_f - \delta_b$ measures the extent of forward-looking behavior versus habits.
- τ: elasticity of intertemporal substitution.
- Can be estimated in a similar way to the NKPC (problem set).

What did we learn?

- IV regression with HAC standard errors
- Weak instruments
- Applications: NKPC, consumption Euler equation

Looking ahead

- How can we estimate the causal effects of economic policies or changes to economic fundamentals?
- Do we always need theoretical model equations (like the NKPC), or can we do causal inference with fewer theoretical assumptions?